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AIoT-enabled digital twin system for smart tunnel fire safety management

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ABSTRACT

High traffic flow in a confined tunnel makes fire safety a critical issue. This paper proposed a digital twin framework for tunnel fire safety management in real-time, driven by dynamic sensor data and AIoT technologies. A deep learning model trained by the Transformer network and simulation dataset is used to predict real-time fire location and size. Then, the AI model is integrated into a 3D digital twin platform developed by the game engine Unity 3D. The performance of the proposed digital twin framework is demonstrated using numerical experiments and large-scale tunnel fire tests. Results show that the established AI model achieved promising accuracy in predicting fire location and power for both numerical and experimental data. The digital twin platform can also visualize the 3D fire scene that supports evacuation, firefighting, and emergency rescue. This research demonstrates the feasibility of using a 3D environment and digital twin in real-time fire safety management.

1. Introduction

Driven by economic and urban development, a large number of tunnels and underground facilities have been built to facilitate transportation. However, the tunnel has an enclosed environment. In a tunnel fire incident, it is difficult for people to evacuate from the narrow space with a single transferring direction (Ingason et al., 2015). For example, a tunnel fire occurred in the Shanxi Yanhou tunnel in 2014, 40 people were killed, and 12 people were injured (Zhang and Huang, 2022). Thus, fire safety management and emergency response for tunnels are essential.

Tunnel fire safety can be improved through several aspects, such as enhancing the safety education of drivers to minimize traffic accidents, optimizing the fire safety design of tunnels, and improving emergency response strategies. However, from the principle of safety management, some accident like collision and fire in a tunnel is inevitable and unpredictable. Thus, to achieve proper fire safety, we need to design a tunnel that can handle any possible fire scenarios (Zhang et al., 2019, 2021a) and improve emergency response to minimize fire casualties and damage through advanced technologies (Ma and Wu, 2020; Wang et al., 2015). Such a smart firefighting system can recognize and quantify fire timely and accurately using neural networks and provide more fire management functions and visualization tools by combining digital twin technologies (Huang et al., 2022).

To improve the fire safety of tunnel and underground space and handle tunnel fires caused by vehicle collisions, several studies have been done to investigate the application of the latest information technologies and computational methods in tunnel firefighting. For example, Wu et al. (2021) demonstrated the possibility of the Artificial Intelligence (AI) method to detect and determine fire sources and hazards. An intelligent tunnel firefighting system based on IoT sensors and AI methods is further proposed to identify fire scenarios in a real-time manner and demonstrated in a reduced-scale tunnel model (Wu et al., 2022a). Guo et al. (2022) investigated tunnel fire scenario reconstruction and key fire parameters prediction by integrating tunnel fire information and machine learning methods. Previous studies showed that it is very useful to deal with tunnel fire problems using smart algorithms

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Nomenclature

Symbols

D	equivalent diameter (m)
H	tunnel height (m)
H_f	flame height (m)
ΔH	heat of combustion (MJ/kg)
L_f	flame length (m)
$\dot{m}$	mass loss rate (kg/s)
Q	heat release rate (kW)
R^2	coefficient of determination
T_f	temperature of the flame tip (K)
T_0	ambient temperature (K)
u_0	ventilation velocity (m/s)

Greeks

η	combustion efficiency

Abbreviations

AI	artificial intelligence
AIoT	artificial intelligence of things
ANN	artificial neural networks
AR	augmented reality
BIM	building information modelling
CFD	computational fluid dynamics
COAP	constrained application protocol
CCTV	closed-circuit television
CV	computer vision
DBMS	database management systems
FDS	fire dynamics simulator
FE	finite element
HRR	heat release rate
HTTP	the hypertext transfer protocol
IDE	integrated development environment
IoT	Internet of Things
ML	machine learning
MLR	mass loss rate
MQTT	message queuing telemetry transport
MR	mixed reality
MSE	mean-squared error
RDTS	raman-based distributed temperature sensors
RNN	recurrent neural network
SQL	structured query language
TCNN	transpose convolutional neural network
VR	virtual reality
WebGL	web graphics library
WSN	wireless sensor networks

and systems.

To enhance fire safety management and firefighting with greater intelligence and reliability, it is crucial to accurately predict fire development and share real-time information about onsite situations and resources. Digital Twin offers a promising solution by providing an executable and dynamic virtual representation of real-world objects or activities, which reflects the geometrical or semantic properties and behavior patterns of objects using the latest data communication and modeling techniques (Qi et al., 2021). A digital twin system consists of several key components, including physical entities, virtual models, services, fusion data, and connections. These components work together to enable real-time integration and interoperation among physical entities, virtual models, and services through fusion data (Glaessgen and Stargel, 2012).

The framework of a digital twin system comprises five essential components.

(1) Physical entities, representing objects with specific functions within the real-world environment, serve as the foundation for the digital representation created by the digital twin.
(2) Virtual models, the digital equivalents of physical entities, replicate the geometrical or semantic properties and behavior patterns of their real-world counterparts.
(3) Service layer, encompassing functions such as management, control, and optimization provided for stakeholders. This enable stakeholders to interact with and utilize the digital twin system.
(4) Fusion data, acting as the core driver of the digital twin, integrating information from various sources to provide a comprehensive and accurate representation of the real-world environment within the digital twin system.
(5) Connection component, facilitating real-time integration and interoperation among physical entities, virtual models, and services through fusion data. This ensures seamless operation of the digital twin system, providing timely and relevant information to stakeholders.

To enhance the fire safety of buildings or infrastructures like tunnels, integrating digital twins with physical facilities or developing a digital twin system integrated with current facility management systems is a feasible approach. This integration enables real-time monitoring, accurate prediction, and effective management of fire incidents, ultimately improving overall fire safety outcomes.

However, several challenges persist in designing and developing a digital twin-enabled safety management system for smart tunnels. Firstly, the deployment of IoT devices necessary for real-time data collection in digital twin systems entails significant investment, leading to low cost-efficiency, especially concerning fire emergency management alone. Addressing this challenge involves expanding the functionalities of the tunnel digital twin system while enhancing its cost-efficiency. Secondly, ensuring adequate coverage of critical fire information poses a challenge, particularly in long tunnels where sensors located far from the fire may not receive sufficient data. Prioritizing sensors closer to the fire source becomes essential to overcome this challenge effectively. Additionally, current deep learning models used in fire prediction may not promptly process the limited data collected during fire situations. Developing and applying new deep learning models to enhance fire information prediction based on the existing sensor deployment is another pressing issue. Thirdly, previous experiments on fire prediction models have primarily relied on numerical experiments and scale-down demonstration simulations, lacking full-scale and realistic fire scenarios. Evaluating the prediction accuracy, system stability, and cost-efficiency of digital twin firefighting systems under such conditions remains a crucial yet unaddressed aspect. These challenges underscore the necessity for further research to enhance the scientific rigor and effectiveness of digital twin-enabled safety management systems for smart tunnels.

This paper presents a novel artificial intelligence of things (AIoT)-enabled digital twin system designed for intelligent tunnel fire safety management. The system leverages advanced deep learning algorithms and numerical simulations to accurately identify and reconstruct fire scenarios in real-time, facilitating swift decision-making during emergencies. The reliability and effectiveness of AI prediction model are rigorously validated through extensive large-scale tunnel fire tests, ensuring its practical applicability in real-world scenarios. This validation process underscores the capability of the digital twin system to provide precise and actionable insights into tunnel fire incidents,

enhancing emergency response efforts. Furthermore, the integration of an immersive fire monitoring digital twin platform in Unity 3D enhances the system functionality by providing stakeholders with a user-friendly interface for visualizing fire scenarios. This platform empowers stakeholders to effectively comprehend and respond to fire incidents, thereby enhancing overall tunnel fire safety management. In summary, the proposed AIoT-enabled digital twin system represents a significant advancement in tunnel fire safety management, offering real-time monitoring, accurate prediction, and immersive visualization capabilities. Its integration of cutting-edge technologies promises to revolutionize the monitoring, management, and response to tunnel fire incidents, ultimately improving safety outcomes in tunnel environments.

2. Key concepts

2.1. Digital twin

The concept of the digital twin was introduced in 2003 by Grieves for product lifecycle management, comprising physical products, virtual products, and their connections (Grieves, 2014). Revisited by NASA in 2012, the digital twin represents the state of a corresponding entity based on historical data, real-time sensor data, and a physical model twin. Described by Gabor, a digital twin is a specialized simulation constructed with expert knowledge and real data, achieving more accurate simulations across different scales of time and space (Gabor et al., 2016). According to Maurer, a digital twin is a representation that depicts production processes and product performance (Maurer, 2017). While cyber–physical system (CPS) and digital twin share conceptual commonalities, their emphases diverge. While CPS and digital twins share conceptual commonalities, they diverge in emphasis. CPS centers on the 3C concept (computing, communication, and control), whereas digital twins prioritize data and models for real-time interoperation between physical entities and their virtual counterparts (Tao et al., 2019a; Jiang et al., 2023a). The concept of the digital twin comprises two facets: the physical twin and its corresponding digital counterpart. Physical entities, augmented with smart devices, update sensing data to cyberspace (P2C), which in turn provides control signals to optimize the behaviors of real-life objects (C2P). Over the past few years, digital twin has been successfully applied in many fields, such as agriculture, construction, manufacturing, smart city, health care, and supply chain, as shown in Fig. 1.

Digital twins can reflect the multi-dimensional attributes and behaviors using models and data and further provide information services such as real-time monitoring, dynamic control, and information sharing (Tao et al., 2019b; Tao and Zhang, 2017). It functions like Building Information Modelling (BIM) to encompass procedures, technologies, and data schemas about the building's physical and functional characteristics. But digital twin integrates more technologies to achieve process-oriented and spatial-temporally dynamic mapping of resources and activities, forming a close-loop control system (Deng et al., 2021). Digital twins can automatically update themselves following the continuous changes of physical twins to realize real-time cyber-physical interoperation and synchronization (Jiang et al., 2022; Su et al., 2022).

For civil and building engineering, Construction Digital Twin (CDT) is a proposed socio-technical mapping of building components that aids collaborative management (Boje et al., 2020). Digital Twin Construction (DTC) is also introduced to utilize real-time monitoring data and AI to optimize construction (Sacks et al., 2020). DTC integrates data clustering and the PDCA cycle for lean planning. While CDT assists construction execution, DTC offers a comprehensive model. Five layers include data acquisition, transmission, digital modeling, integration, and services. Various technologies are applied for data collection, fusion, and visualization (Tuhaise et al., 2023).

For building fire safety, Digital Twin is still a relatively new concept with much fewer publications (Wu et al., 2022a; Zhang et al., 2024). A Digital Twin-based framework is proposed to improve the capability of indoor safety management, which combines the Internet of Things (IoT), building information modeling (BIM), and machine learning methods (Liu et al., 2020). Zhang et al. (2022a) developed an Artificial-Intelligence Digital Fire (AID-Fire) system for real-time building fire identification, and such a digital twin system was demonstrated with real-scale compartment fire demonstration. The trained AI model can automatically process the obtained sensor data, and real-time 2D scene and critical fire information (e.g., fire power and location) can be displayed. Data communication between digital and physical is a key

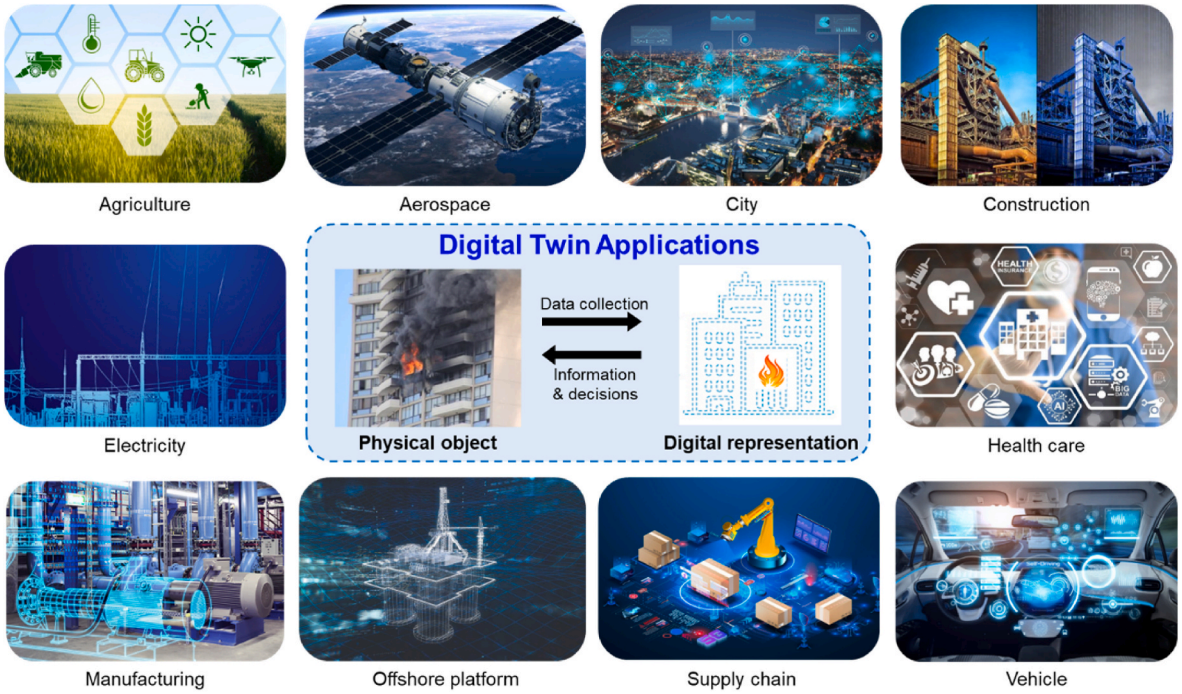


Fig. 1. An overview of digital twin systems and their application fiels.

issue in digital twin development, semantic web technologies are usually adopted to solve the challenge of operability between BIM and IoT data of DT-based applications (Jiang et al., 2023b).

2.2. Smart firefighting

In building fire applications, AI, Virtual Reality (VR), IoT, and BIM are usually the key components to establishing a digital twin, and there has been an increasing number of related studies conducted in recent years. AI model trained with massive data (e.g., simulation, experimental data, or historical data) can be used to predict real-time fire information (e.g., fire size or location) or forecast future events (e.g., flashover, backflow, or structure response) in advance. A data-driven method was adopted to predict the temperature and velocity profile inside the compartment (Hodges et al., 2019). Real-time prediction of fire heat release rate (HRR) using deep learning algorithms and flame images has been investigated (Wang et al., 2023), and transient fire HRR can be well recognized by the proposed method.

Because of the challenges in conducting massive large-scale fire tests and form an experimental database, coupling ML and numerical fire simulations were more widely used to train the prediction model. A machine learning forecasting model P-Flash was developed to predict the risk of flashover in multiple-room compartment fire (Wang et al., 2021a). Heat detector temperature data generated from one thousand numerical simulations were used to train the model. Smoke movement during tunnel fires (i.e., backflow occurrence and back-layering length) can also be predicted by machine learning methods in previous study (Hong et al., 2022). Their results showed that the ANN model has the best performance to estimate the back-layering length. A framework based on Finite Element and machine learning was developed to predict the real-time structural response under fire (Ye et al., 2022a). Temperature data and structure displacement from the numerical simulation are used as input and output to train the ML mode.

Building information modeling (BIM) and related technologies are usually used to generate and manage the digital representation of physical assets, which can be utilized to develop Digital Twins. Their applications in fire safety management have been widely investigated (Li et al., 2018). A BIM-based Intelligent Fire Prevention and Disaster Relief System was developed to support early fire detection and fire rescue (Cheng et al., 2017). Bluetooth-based sensors were used to monitor the fire information and show the real-time fire information on mobile devices. BIM is also applied in emergency evacuation design, a fire evacuation model was developed to facilitate the two-way data exchange in fire evacuation design (Yakhou et al., 2022). BIM-based indoor localization method was investigated to support the emergency response during fire events (Li et al., 2014). The localization accuracy and robustness of the proposed algorithm were proved via serval simulation fire scenarios.

BIM related technologies are also adopted to study the behavior decisions of building users during evacuation. A fire emergency management system with the function of intelligent fire monitoring and optimal evacuation route planning was developed to study user decisions (Ma and Wu, 2020). Results show that the evacuation effectiveness of the proposed method is better than the evacuation process without auxiliary guidance measures. Another research found that the evacuees' decisions during the emergency will be affected by their neighbors' decisions (Fu et al., 2021). Evacuees tend to follow others when making risky decisions. The evacuation during an emergency can be investigated or optimized by integrating BIM and VR (Ronchi et al., 2015; Arias et al., 2019). A VR model developed from BIM was applied to investigate the factors of evacuation decisions (Lovreglio et al., 2022). They found that both unfamiliarity and the presence of smoke negatively affect the evacuation choice. BIM technology has also been applied in fire safety design (Siddiqui et al., 2021; Tang et al., 2021) and fire risk assessment (Wang et al., 2021b; Hosseini and Maghrebi, 2021).

To address the above challenges, this work presents a digital twin framework for the simulation and connection between real tunnel and virtual tunnel model, especially during tunnel fire events. The remainder of this paper is organized as follows. This work is going to present a framework for the simulation and connection between the real tunnel and virtual tunnel model, especially during tunnel fire events. The second chapter will introduce the overall framework and each component of the proposed system, and the third section will build an AIoT-enabled immersive digital twin for real-time fire recognition. Then, some demonstrations and discussions will be given in the fourth section. The final chapter will conclude the present work.

3. Digital twin framework for tunnel fire safety management

A digital twin is regarded as a virtual to reflect the physical object in Grieves's whitepaper (Grieves, 2014). It consists of three main parts, according to his whitepaper: (1) physical entities (or assets) in the real world, (2) virtual models in the digital world, and (3) real-time integration and interoperation between the physical world and the digital world.

Fig. 2 demonstrates the overall framework for the proposed tunnel safety management digital twin, including the components (i.e., physical tunnel, digital tunnel, data management platform) and their

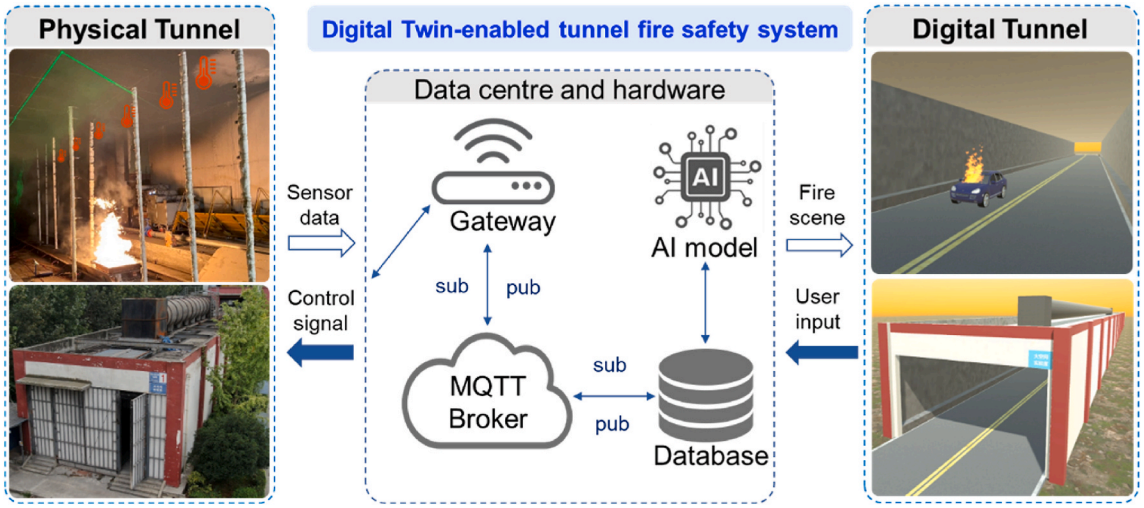


Fig. 2. The overall Digital Twin framework for tunnel fire safety management.

relationships. The real tunnel is the physical entity in the Digital Twin model, and the digital tunnel is the virtual model. The data and information between the real tunnel and virtual tunnel are the connection between physical and virtual spaces. Compared with our previous system (Wu et al., 2022a), this Digital Twin not only can recognize and forecast real-time tunnel fire but also support daily facility management and fire safety emergency response. The Digital Twin model can be integrated with other facility management systems, such as tunnel ventilation and lighting systems, to control their operation and urgent use during a fire emergency.

The data transfer between the physical tunnel and the digital tunnel has to be bidirectional; otherwise, it is only a Digital Shadow rather than a Digital Twin (Ma and Wu, 2020). From left to right, the data is transferred from the physical tunnel to the digital tunnel, and the real fire scenario is projected into the digital space. At the same time, the data (e.g., the facility control signal or evacuation/rescue instructions) can be transmitted from the digital tunnel to the physical tunnel. For example, the user can operate in the digital tunnel to control the smoke ventilation and fire suppression system in the physical tunnel.

3.1. AIoT-enabled physical infrastructure

In the proposed framework, the tunnel itself and related physical entities constitute the physical space. Three categories of entities are defined, including fixed entities (e.g., tunnel systems), movable entities (e.g., passing vehicles), and probabilistic entities (e.g., accidents and fire events). The tunnel itself usually does not change over time, except in some extreme situations, such as structural collapses due to severe fire or vehicle collisions. Tunnel structure resilience under fire is important for tunnel safety, and many previous studies have explored the behaviors of tunnel structures during tunnel fire (Ye et al., 2022b). Due to the extremely low probability of occurrence, the structural failure of the tunnel is not considered, and the tunnel itself is considered an immutable object and the container of the digital twin in the proposed framework.

The facility management system is used to maintain the daily operations of the road tunnel, such as the power supply, lighting and waymarking systems, or used to control or mitigate various risks, such as drainage, firefighting piping, and smoke ventilation system. In the current study, the proposed digital twin system aims to manage fire risk and guide firefighting activities. Therefore, only the facility management systems related to fire safety are included in the physical layer. The details of the facility management systems are listed in Table 1. These conventional physical systems could update the real-time information (e.g., location, ID, status, data flow, and) to their corresponding digital twins or be automatically controlled or through visual input.

The passing vehicles or passengers belong to the movable entities of the system, which may control or be affected by fixed entities. During normal operation of the tunnel, vehicles only pass through the tunnel quickly. During the emergency, the vehicles may cause or be stopped by accidents or fires, and passengers may need to be evacuated from the tunnel. Then passengers or vehicles may interact with the fixed entities. For example, the vehicles exited the tunnel according to the prompt of the lane control light, and the trapped people evacuated from the tunnel followed the evacuation sign. Unlike industrial or construction situations, vehicles or passengers cannot be traced using installed or wearable sensors. With the recent advancement of deep learning techniques and computational capabilities, humans and vehicles can be identified and tracked, and the speed can be estimated using computer vision methods and surveillance videos (Hua et al., 2018). However, the investigation of object tracking under fire and smoke environments is still limited. The integration and application of millimeter wave, LIDAR, Ultrasound, and Infrared cameras may help to identify the object under low-visibility environments (Starr and Lattimer, 2014).

Table 1
Facility management systems related to tunnel fire safety.

System	Functions
Sensor network	Monitor and collect environmental information (e.g., temperature, humidity, smoke, etc.)
CCTV	Transmit visual information using video cameras
Communication	Transmit the signal wirelessly (e.g., Zigbee, 4G, and LoRa) or by wire (e.g., RS485 or PLC)
Smoke ventilation	Provide fresh air and exhaust out harmful gases (e.g., jet fans or exhaust fans)
Emergency lighting	Facilitate the evacuation in emergencies such as fire
Lane control	Manage vehicles on a multi-way road with light or sign
Fire suppression	Detect, control, distinguish or even prevent the fire
Warning board	Provide warning information in an emergency
Evacuation sign	Guide the evacuation to safe places using static or dynamic signs

The fire hazard can be quantified by the probability of occurrence before the fire and the severity after the fire. Before the fire, the vehicle type can be identified, and the motion can be tracked using computer vision methods. Each vehicle can be marked with different fire risk levels according to its vehicle type, size, and speed. The overall tunnel fire risk can be derived by considering the risk of each vehicle, traffic flow, and the tunnel environment (e.g., lighting, tunnel geometry, weather, etc.). The severity of tunnel fires can be evaluated by their size (or power), location, visibility, distribution of temperature and toxic gases, spread speed, etc. The fixed entities in tunnels can be used to detect, monitor, and distinguish the fire, and guide the trapped people to evacuate. For example, the fire can be detected by sensors and be monitored by surveillance cameras, and distinguished by sprinkler system. Then, people can evacuate to safe places by following the dynamic evaluation or warning signs.

3.2. Real-time cyber-physical interoperation

The fire safety digital twin system in the current work has the capability to evolve alongside the dynamic tunnel environment by utilizing collected data from sensors or cameras. This comprehensive tunnel fire digital twin encompasses data management processes including acquisition, processing, and visualization, as demonstrated in Fig. 3.

In the proposed tunnel safety digital twin system, the data could be collected from the IoT sensors, cameras, and other facilities inside the tunnel, or data generated by simulation or AI. Sensor data mainly include environmental parameters such as temperature, humidity, and gas concentration. These parameters are important for identifying fire occurrence and predicting fire development. The camera data is useful for monitoring the fire risk and detecting fire with the help of computer vision technology. However, the camera will be quickly blocked by high-density smoke after a fire occurs. Therefore, the combination of sensor and camera data is more reliable. The tunnel facilities, such as the ventilation system, fire protection system, lighting system, and evacuation facilities, are helpful in responding to fires or accidents. Therefore, it is necessary to monitor the operation statuses of these facilities.

The collected data could be transmitted either through wired or wireless means. To ensure the reliability of the system, wired data transmission is better, as the data transfer speed and capacity are sufficient, but the wired sensor may cost more money and be difficult to update after the tunnel is built. Wireless sensor networks are more and more popular due to their high flexibility and convenience. In fire scenario, both wired and wireless sensors may be quickly destroyed by fire, resulting in the system being unable to complete subsequent fire monitoring and prediction functions. Therefore, it is important to enhance the system's reliability. Some approached can be adopted to address the realiablity issues of sensors, including sensor redundancy, fire-resistant materials, advanced transmission technologies, and regular facility mantance and testing. For long tunnels, a large number of gateways may be needed for a short-range network like Wi-Fi and Bluetooth. For the data transmission protocols, MQTT, HTTP, and COAP

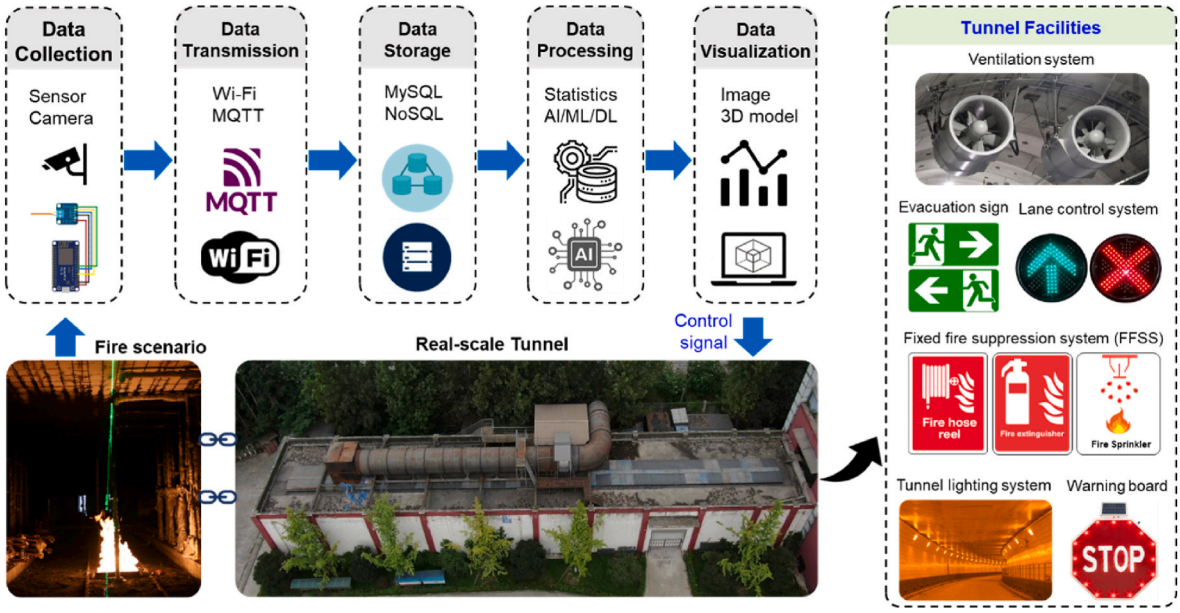


Fig. 3. Data management and interoperation of tunnel fire digital twin system.

are the most widely used IoT technologies.

After the data is collected, it needs to be well stored before further processing and management. Database techniques are the basis for data storage. Although some new data storage methods, such as NoSQL and cloud storage, have been developed recently to handle the massive amount of data, the traditional relational database is still very popular. Structured Query Language (SQL) and relational database management systems (DBMS) like MySQL are widely used to manage data in different applications and platforms. In the present work, the MySQL database is used to store the sensor and simulation data.

Data processing aims to provide valuable information and support critical decision-making using IoT data. In the proposed system, the raw data mainly include environmental sensor data, visualization data, and system operation data. The first step is to preprocess the data, like data cleaning, data compression, and reduction. Then, data analyzing methods, including statistical methods and neural network methods, can be utilized to further process the preprocessed data. This work analyzes the temperature data collected from sensor networks by the deep neural network to predict fire information like fire size and location. The system operation data can be processed using some simple statistical methods to monitor the real-time object status. Data fusion with multiple features or data sources, such as CCTV cameras, and smoke and temperature sensors, can be adopted by mixing the data in mathematical models to better understand the real fire situation.

After processing the data and obtaining the desired results, visualization technology is required to display the results. Visual analytics aims not only to view the data from different perspectives through data mining, statistics, and visualization tools but also to enable human interaction (Protopsaltis et al., 2020). Currently, different types of charts, 2D/3D rendered graphics, VR/AR/MR, and Digital Twin are the most widely used visualization methods. In the proposed system, an interactive user interface, including 3D geometry and real-time charts, is provided to display intuitive onsite fire information and warnings of critical fire events. The tunnel facilities related to fire safety can also be controlled through the user interface by integrating the Digital Twin system and the facility control systems.

3.3. *Functional digital twin for tunnel firefighting*

The proposed digital twin system is designed for managing tunnel fire safety for its whole lifecycle and different situations. The functions of the system focus on four aspects, monitoring and prediction of tunnel fire risk, control and management of tunnel facilities, fire identification and forecasting, and emergency evacuation guidance, which cover normal situations, fire situations, and emergency evacuation situations, respectively. The detailed descriptions of these functions are illustrated as follows and are shown in Fig. 4.

(1) **Daily tunnel transportation and fire risk monitoring.** The first function aims to monitor the vehicles and eliminate the fire risk prior to the fire. In view of safety science, the actions taken before the accidents are most useful, although these measures are often overlooked. From the perspective of fire safety management, there are many rules or guidelines to identify and eliminate fire risk, but these procedures are highly dependent on the operations of related personnel, and human failure may occur even if they are well trained. As a comparison, a machine with advanced algorithms and techniques can avoid human errors. It is more achievable and reliable to monitor and reduce fire risk using technology. In the proposed system, the heat or temperature sensors as well as the CCTV cameras, are used to monitor the fire risk, as they have already been installed in the most important transportation tunnels.

(2) **Tunnel facilities management and control.** Facilities related to fire safety inside the tunnel should be well-managed. During the daily operation, the facilities' statuses should be monitored, inspected, or maintained regularly, and these facility management data shall be updated in time and shared with the fire safety digital twin system. During a fire, these fire safety facilities shall be activated to extinguish the fire or to mitigate the fire risk before the fire. For example, the corresponding ventilation and fire protection facilities near the fire shall be automatically activated based on the detected fire information to exhaust toxic smoke or put out the fire. These systems can also be manually controlled, which is very useful during emergencies or under a system error. All mentioned information can be integrated into the proposed digital twin system to enable better tunnel fire safety management.

(3) **Fire identification and prediction.** The digital twin system should provide helpful fire information to help tunnel managers or the fire department make better decisions. In case of a tunnel fire, a proper response during the fire can minimize casualties and

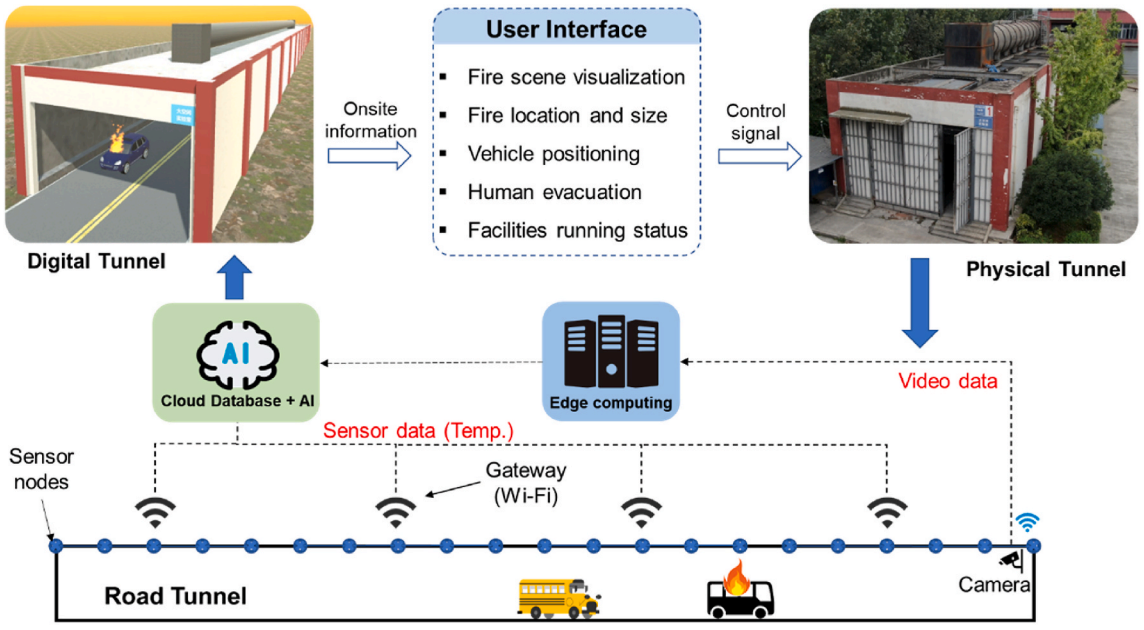


Fig. 4. Functions and data flow of the digital twin system for firefighting.

save lives. The critical fire information shall be identified in real-time or be predicted in advance, such as the fire location, fire size (HRR), tendency of fire spread and growth, distribution of temperature, visibility, and toxic gases. The fire information can be obtained by the sensors or devices installed inside the tunnel. For example, fire can be detected and localized using the conventional smoke and heat detector network or the latest fiber-optic temperature measurement (Ishii et al., 1997). However, the estimation accuracy of such a temperature sensing system may be easily affected by the environmental conditions through influencing the temperature distribution, such as the wind flow. As a comparison, using machine learning to identify and predict tunnel fires can give more comprehensive and accurate results.

(4) **Evacuation and rescue guidance.** The system will also provide guidance for evacuation and rescue activities during emergencies like fire events. The rescue under extremely low visibility environments is challenging, especially the long tunnel in suburban areas. It may take a considerable amount of time from the time stamp when the fire is detected to the time stamp when firefighters arrive at the fire site (Zhang et al., 2023). During the fire, the trapped occupants are at great risk and need to evacuate as soon as possible. The only information that can be easily obtained by evacuees to help with their evacuations are the instructions from the technical installations of the tunnel, such as the information signs, lighting, and alarms. These installations have been proven to significantly affect the evacuation performance of evacuees during tunnel fires (Zhang et al., 2021b). The proposed digital twin system aims to deliver useful onsite information to guide the evacuation process using technical installations or facilities. These facilities can be integrated with the digital twin system, and the dynamic information may be given based on real-time fire information and other conditions. For example, dynamic evacuation signs based on real-time fire information may have a higher probability of helping evacuees survive than static signs.

4. AIoT-enabled fire cognitive system

Among these functions proposed above, real-time fire recognition and rendering are the most crucial during a fire, such as the fire size and fire location. Therefore we will focus on the development of a fire recognition model and real-time fire rendering platform. The fire recognition model is developed based on a deep learning algorithm and a large amount of data. The data used in the present work is generated by numerical simulation due to insufficient fire data from real fire or experiments. This section will describe how the training dataset is generated, how the AI architecture is built, and how the platform is developed.

4.1. Dataset generation from numerical simulations

The aim of the proposed fire recognition model is to identify the tunnel fire scenario in real-time, including tunnel fire location and HRR (or fire power), which requires a comprehensive tunnel fire dataset. To generate enough data for training, numerical simulation is the most cost-effective way. Many studies have conducted numerical simulations to study the tunnel fire behavior or key parameters, such as the smoke movement (Zhang et al., 2021a; Hu et al., 2007), ventilation (Verda et al., 2021), and the tunnel ceiling temperature distribution (Ji et al., 2015; Huang et al., 2018). Compared with the limited experimental data, numerical simulation can obtain more detailed results (e.g., temperature and velocity contour), and these detailed and comprehensive data can be used to get conclusions that cannot be found from the experimental studies. Using Computational Fluid Dynamics (CFD) simulations to generate training datasets for fire prediction model training has also been widely demonstrated (Hodges et al., 2019; Wu et al., 2022b).

In the present work, the Fire Dynamics Simulator (FDS) version 6.7 released by NIST is used to simulate different tunnel fire scenarios and generate training datasets. A 3D full-scale tunnel model is established according to the laboratory tunnel in Sichuan Fire Research Institute (SCFRI), which has a dimension of 140 m, 8 m, and 6 m in length, width, and height, respectively. Ventilation is not considered in the simulation. Both portals of the tunnel are set to "Open" and connected to the ambient. A square "Burner" with n-heptane fuel is adopted as the fire source. The fire is located at the center of the transverse direction but at different longitudinal locations in different scenarios. The constant HRRPUA value of 1000 kW/m^2 is adopted, so the area of the fire source controls its HRR or power.

Fire location and fire size (HRR) are two key parameters to

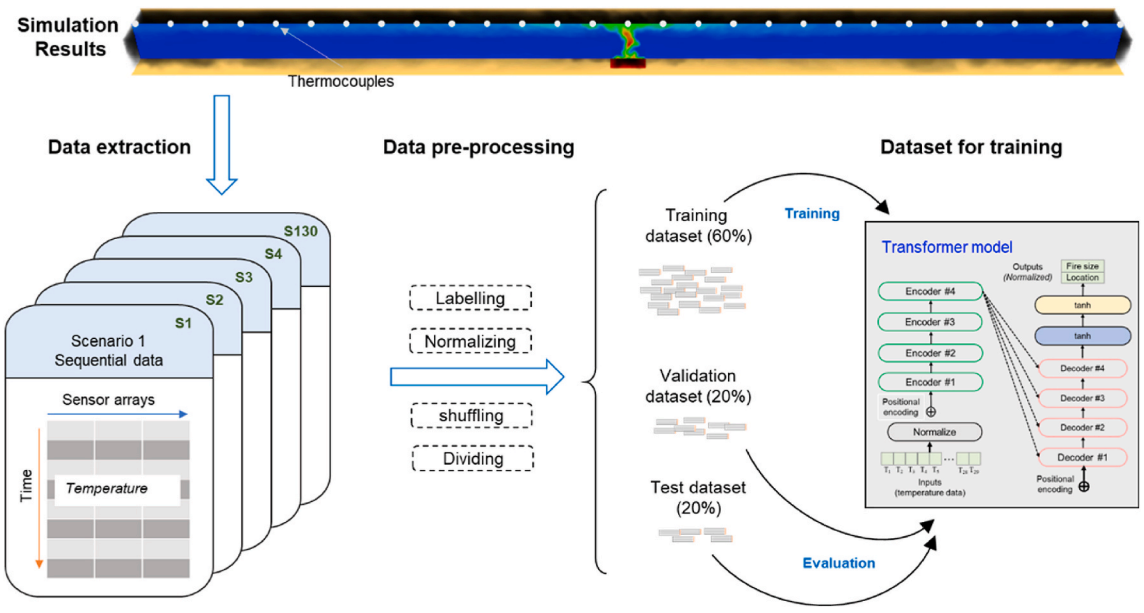


Fig. 5. Generation of the database for model training.

characterize tunnel fire behavior, and they were varied to simulate different fire scenarios. From the tunnel's left entrance, 13 different locations from 10 m to 130 m with an interval of 10 m are selected for the simulation. Ten different HRR values 0.16, 0.36, 0.64, 1, 2.25, 4, 9, 16, 25, and 36 MW, were chosen to model different types of fires. Other simulation conditions and procedures are the same as in previous work (Zhang et al., 2022b). Compared with previous work, smaller fire sizes are also considered, and more accurate results could be obtained by AI prediction. A similar mesh sensitivity checking process is conducted to evaluate the grid resolution (Zhang et al., 2022b). The grid size of 0.1 m is selected for all the simulations in the present work.

To capture the real-time gas temperature, 29 thermocouple devices were mounted on the tunnel ceiling with an interval of 5 m along the centerline of the tunnel. A temperature contour at the tunnel center plane is defined to record and visualize the tunnel contour animation. The simulation time for all scenarios is last for 120 s to get to the convergence state. After the simulation, the results are obtained and preprocessed to form the training dataset. Fig. 5 demonstrates the process of database generation from numerical simulations. The temperature data and corresponding fire information data for different scenarios are extracted to form a 15,600 (130 scenarios × 120 s) training dataset.

4.2. A deep learning algorithm for fire recognition

For tunnel fire safety, deep learning methods have been used to identify the fire source (Wu et al., 2021), predict the critical ventilation velocity (Zhang et al., 2021a) and the real-time temperature distribution (Wu et al., 2022b), and forecast the transient fire scenarios (Zhang et al., 2022b). In the proposed digital twin system, sensor data is input into the system, and fire scenarios (e.g., severity and location) could be generated. Due to the correlation between input and output is hard to acquire through statistical methods or empirical equations, the AI model is used to conduct the transformation between input and output. When a fire occurs, both sensors and cameras can collect the information, but cameras will soon be obscured by high-density smoke and then cannot get any useful information. By contrast, sensor data becomes more valuable in such a situation. Therefore, sensor data is more reliable and preferable for fire recognition. In the present work, a Transformer-based deep learning architecture is used to identify the real-time tunnel fire scenarios using temperature data (Li et al., 2023).

Transformer is designed to process sequential data and has been widely adopted in language translation (Devlin et al., 2018) and computer vision (CV) (Dosovitskiy et al., 2020). It is a deep learning architecture based on the self-attention mechanism (Vaswani et al., 2017). Compared with recurrent neural networks (RNN), the Transformer model operates by processing the entire input sequence, and the weights and correlations of each input data are generated using the attention mechanism so that less training time is required. The sensors near the fire source can capture more important information than other sensors, so greater weighs should be assigned. Therefore, the self-attention-based Transformer model might perform better with distributed sensor data. Fig. 6 demonstrates the architecture of the proposed model. The fire information is predicted with input sensor data. Four encoder-decoder layers are adopted to understand and extract information from input data and generate results. In the encoder-decoder layer, multi-head attention structure and dense layers are included. Normalization and inverse normalization functions are added to improve the model performance and boost training speed.

The model's performance was then assessed by utilizing various metrics after training. In Fig. 7, the Mean-Squared Error (MSE) loss and the coefficient of determination R^2 on both training and validation datasets are displayed, which evolve as the training epochs. R^2 can estimate the matching degree of the regressed results and actual values, and MSE can quantify the deviation between the predicted results and actual values. In the present study, predicting fire locations and fire sizes is a typical regression problem, R^2 and MSE loss offer a comprehensive view of the model's performance. Generally, the higher R^2 score of the trained model, the better its performance and prediction ability, and lower loss value means smaller prediction error magnitude. As expected, the loss value on the training dataset decreases steadily with the training epochs. Both training loss and validation loss gradually decrease to relatively low values and stabilize at approximately 0.005. The results suggest that the model convergence is within 200 epochs. In contrast, the R^2 coefficient presents a rapid increase in the beginning and ultimately settles at 96.5 % for both training and validation datasets. The close value of MSE and R^2 on both training and validation is caused by the configuration of the dropout layer to avoid overfitting.

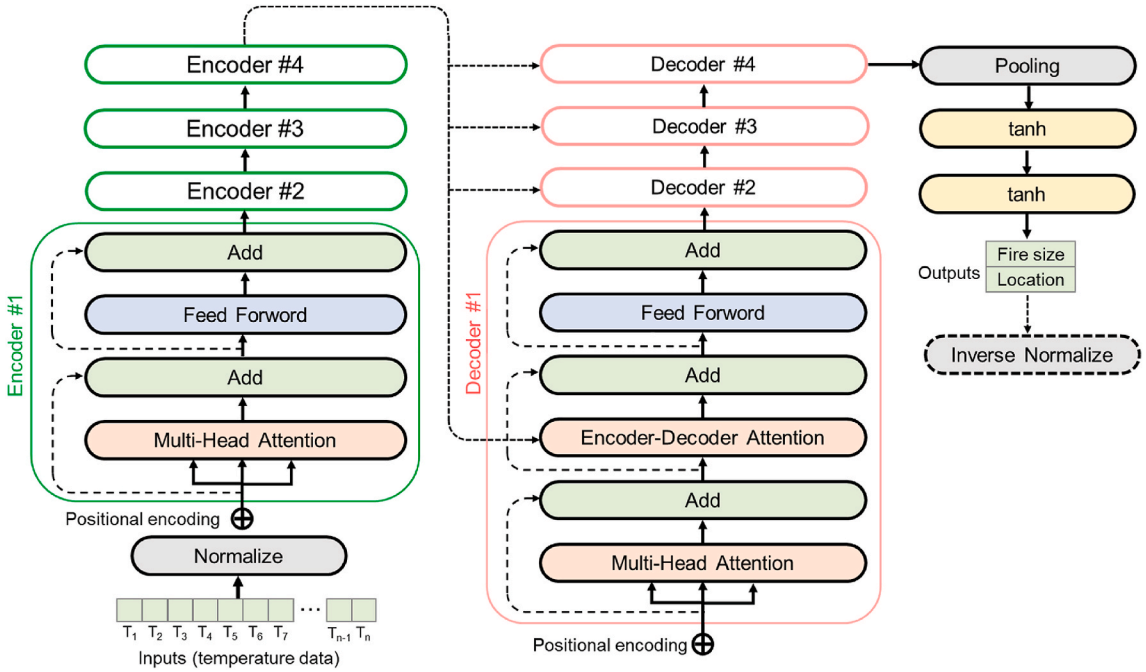


Fig. 6. Architecture of the proposed Transformer model.

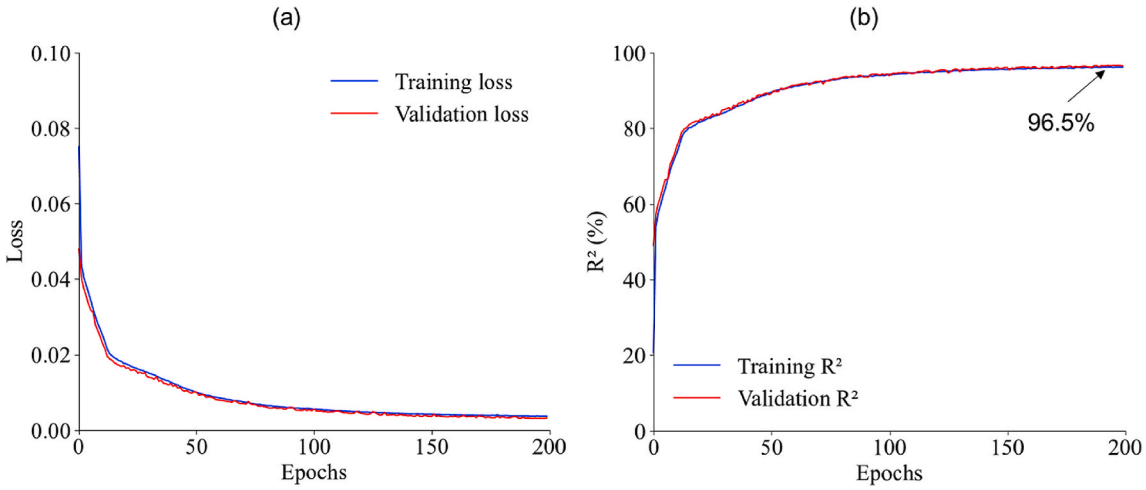


Fig. 7. Evaluation metrics of proposed model: (a) Loss (b) R^2.

4.3. *Immersive fire monitoring digital twins*

This work aims to build an immersive tunnel fire digital twin based on the information of SCFRI tunnel, so the tunnel model has the same dimensions of 140 m × 8 m × 6 m, as compared in Fig. 8. The mentioned tunnel has installed a ventilation system to exhaust the polluted air and smoke generated by fire. Inside the tunnel, there are train platform and an abandoned train carriage. The wall is well protected using fireproof materials.

Based on the proposed framework and trained deep learning fire recognition model, an immersive tunnel fire monitoring digital twin is developed to facilitate fire risk management and real-time firefighting guidance. The digital twin system is built for the SCFRI tunnel with the same dimension and many detailed features, as demonstrated in Fig. 8. The model of a 3D road tunnel and different types of vehicles are constructed in 3D design software *SketchUp* on the same scale, some non-essential components have been simplified to facilitate modeling and real-time rendering. Particle system module in the game engine *Unity 3D* is adopted to create the realistic effects of tunnel fire and smoke movement, which is the state-of-the-art technique to render real-time fire and smoke. The status of flame and smoke can be adjusted based on the predicted fire source. Although the physical properties of the fire and smoke cannot be perfectly simulated in the proposed system, the rendering effect looks extremely realistic, as shown in Fig. 8(e). The functions and logic of the digital twin are developed in *Unity 3D* and converted into a web-based application using Web Graphics Library (WebGL) API to facilitate execution on different platforms.

LAMP stack is one of the most popular open-source software stacks for web development and is used to establish the web-based digital twin platform. LAMP is the acronym for four necessary components to establish a fully functional web development environment: Linux, Apache, MySQL, and PHP. Linux is a free and open-source operating system that serves as an integrated development environment. Apache HTTP Server is used as the webserver to host the WebGL application. MySQL is an open-source database management system to store data. PHP is the programming language that works with web servers and

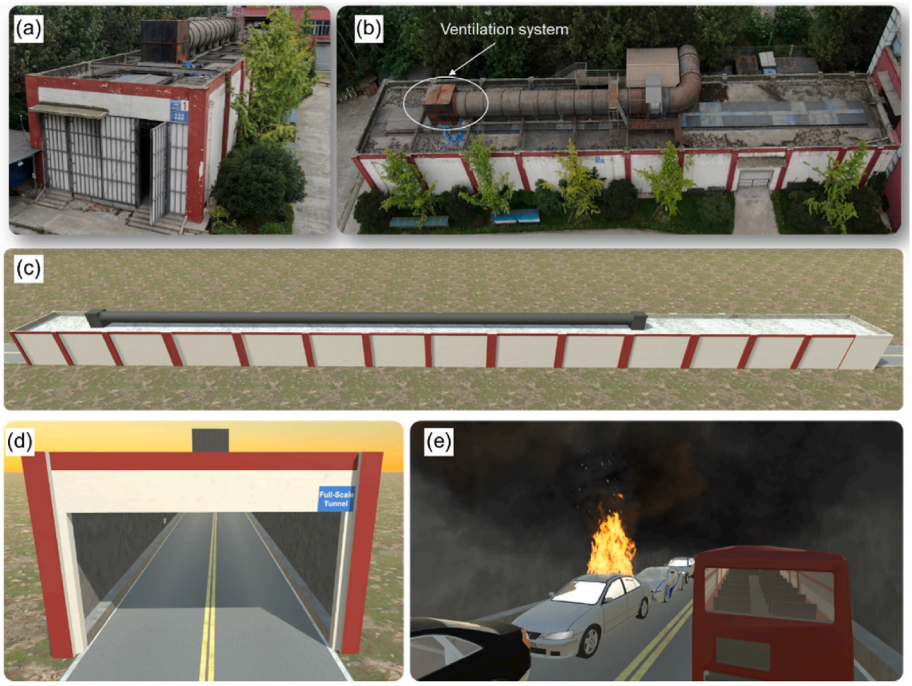


Fig. 8. Tunnel model and fire smoke effect in digital twin system: (a) front view of real tunnel, (b) bird-eye view of physical tunnel, (c) bird-eye of digital tunnel, (d) front view of digital tunnel, and (e) vehicle model, fire and smoke effect inside digital tunnel.

databases for dynamic data processing. Of course, some alternative software can be used to achieve the same function, such as Nginx Server, PostgreSQL Database, and Python scripts.

As the physical space of the digital twin system, the real-scale tunnel deployed Wireless Sensor Networks (WSN) to sense and collect the environmental information inside the tunnel. The adopted wireless temperature sensor consists of four main components: ESP8266 NodeMCU, Type K thermocouple, MAX6675 converter module, and Lithium-ion battery, as shown in Fig. 9. The ESP8266 can be programmed to connect to the Wi-Fi network using Arduino code in Arduino Integrated Development Environment (IDE). Multiple sensors distributed along the tunnel can communicate with the Base Station directly to form a single-hop wireless network. Because the sensors can be destroyed under fire environments, each sensor should work independently. To improve the battery life, the WSN is awaked at fixed intervals when there is no fire, and it will continuously work when a fire is detected.

Message Queuing Telemetry Transport (MQTT) protocol, with publish-subscribe architecture, is adapted to transmit data between WSN and the MQTT server. MQTT is a lightweight protocol that runs over TCP/IP and can achieve bidirectional data interoperation in soft real-time. The real-time temperature data array can be published to the MQTT server by the WSN, and a Python script can be used as the client to subscribe to the specific MQTT topics. The MQTT temperature data can be stored in a Database through Python script or MQTT Rule Engine. MQTT data can also be used to control fire facilities like ventilation and alarm systems by publishing control orders to the devices automatically or manually. After the sensor data is stored in the database, the AI model can read the data from the database and give predictions. The prediction results can also be stored in the database. The WebGL platform can communicate with the database to render the fire and display important information.

The proposed tunnel fire digital twin is to provide an immersive virtual environment for different firefighting purposes, so fire rendering is important. According to Section 3.2, fire location and fire size can be predicted by the trained AI model in real time. The fire size and fire location can be mapped to the digital space (WebGL) using C# language, and corresponding fire scenes can be rendered (e.g., fire dimension, temperature distribution, smoke color, and smoke density). Due to the digital model being designed on a 1:1 scale, the location of the fire size can be easily determined. The fire size or HRR can be represented by the dimension of the fire. The dimensions of the fire, including diameter and height, can be calculated using empirical equations.

Spilt pool fires are common fire sources in the case of tunnel fires, which are resulted by the ignition of leaked liquid fuels after vehicle accidents. The combustion coefficient and the mass burning rate of a spill fire are usually much lower than the corresponding deep pool fire, and the average heat release rate per unit area (HRRPUA) of gasoline or diesel is around 1 MW/m^2 (Li, 2019). Then the burning area and equivalent diameter of the fire source can be estimated from the HRR and HRRPUA. Flame length is another parameter to quantify the fire risk. The flame may impinge on the tunnel ceiling and extends along the tunnel during a large tunnel fire. If the fire size is small and the flame cannot reach the tunnel ceiling, the flame height equals the flame length. The flame length or height of buoyancy-driven turbulent diffusion flames in a no-wind and open environment H_f can be estimated as follows (Heskestad, 2016):

$$H_f = 0.23Q^{2/5} - 1.02D \tag{1}$$

where Q is the HRR (kW), and D is the equivalent diameter of the fire source (m). However, the calculation of flame height inside the tunnel should consider the air flow induced by wind or ventilation system. Besides, Eq. (1) does not apply when fire size is too large to reach the tunnel ceiling.

For the flame length in tunnel fires considering longitudinal ventilation, an empirical equation based on large-scale tunnel fire tests and Alpert's ceiling jet temperatures equation is used to estimate the horizontal flame length L_f (Lönnermark and Ingason, 2006):

$$L_f = \frac{1370Q^{0.8}u_0^{-0.4}}{\left(T_f - T_0\right)^{3/2}H^{3/2}} \tag{2}$$

where u_0 is the longitudinal air flow velocity or ventilation velocity (m/s), T_f is the temperature of the flame tip (K), T_0 is the ambient tem-

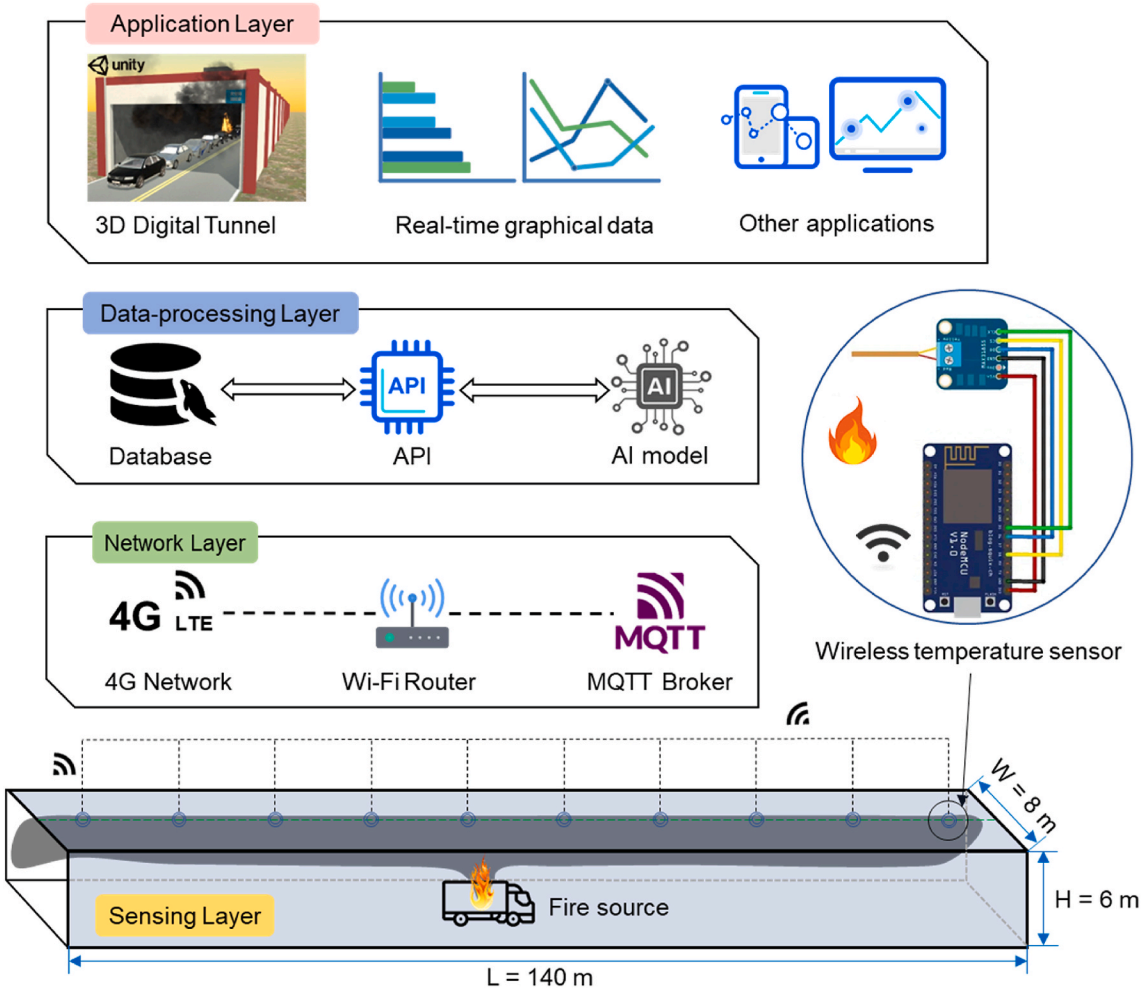


Fig. 9. Implementation of digital twin-enabled smart tunnel fire safety monitoring system.

perature (K), and H is the tunnel height (m). To simplify the fire rendering, the flame height is considered as the tunnel height when the flame reaches the tunnel ceiling.

5. Implementation of digital twin framework

In this section, numerical experiments and two full-scale demonstrations are conducted to evaluate the performance of the proposed Digital Twin system in real-time fire recognition. The numerical experiments are conducted based on the CFD simulation data with different fire scenarios. Then, full-scale demonstrations are conducted based on real-scale tunnel fire test data. The experimental conditions and the fire scenarios will be introduced in the following sections.

5.1. Numerical experiments

Generally, it is difficult and expensive to conduct large-scale fire experiments. Therefore, the established digital twin platform should be first verified using numerical experiments. As an alternative to experimental data, simulation data is input into the system to test and demonstrate the functions and capabilities of the system. Two sets of numerical tests are conducted with changing fire locations and fire sizes, which have not been included in the training dataset of the proposed fire recognition model. In real tunnel fire scenarios, the fire may grow or spread over time. It is significant to predict the transient tunnel fire scenarios.

Firstly, two different tunnel fire scenarios are simulated using the FDS code, as illustrated in the previous section. The tunnel geometry, boundary conditions, and fuel type are kept constant in section 3.1. The fire size (HRR) varies with time in the first fire scenario, which is used to verify the prediction capability of the model and the Digital Twin system. The profile of the simulated fire size is shown in Fig. 10(a). The fire size will grow from zero to the peak value (10 MW), remain relatively stable, and then decrease to zero. The second simulation case aims to demonstrate the ability of real-time fire location prediction, and the curve of fire location versus time can be seen in Fig. 10(b). The fire spread is modeled by a series of striped fire vents. The "CTRL" function in FDS is adopted to sequentially control the state of numerous fire vents (Zhang et al., 2022b). Both scenarios are simulated for 900 s to ensure the development of the fire. During the simulation, ceiling temperature sensor data, fire size, or fire location data are recorded per second.

Then the temperature data are fed into the trained Transformer model to predict the fire size or fire location. The predicted transient fire size or location is displayed in Fig. 9 and compared with the simulated HRR or fire location. For the HRR prediction case, the model can accurately predict fire size changes, demonstrating the trained AI model's capacity to predict transient tunnel fires, as shown in Fig. 10(a). The AI model can also give relatively accurate results for fire spread prediction cases but reflect greater fluctuations, as illustrated in Fig. 10 (b). Such results can be attributed to the simulation settings. It should be noted that the prediction results are poor in the beginning stage for both cases. It is because the simulations have not yet reached a steady state in the beginning, so the obtained temperature data are inaccurate. It is worth noting that the ventilation in tunnel fires is significant, including

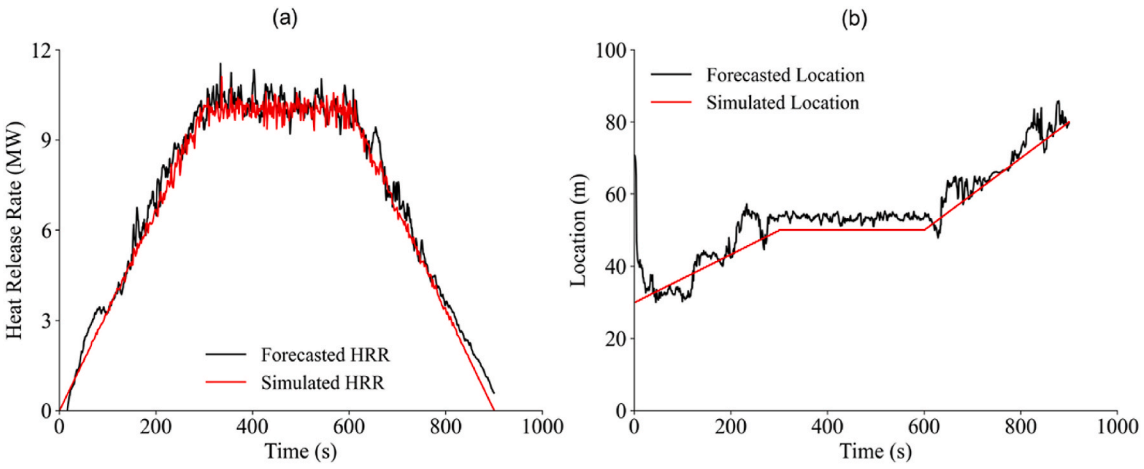


Fig. 10. Comparison between the simulation and prediction: (a) HRR, (b) Fire location.

natural ventilation and mechanical ventilation, fire simulation considering ventilation effect can make the prediction model more adoptable to real fire scenarios.

Finally, the proposed tunnel fire digital twin framework and established fire recognition model are integrated into an immersive tunnel fire safety management platform. The developed platform has five main functional areas, which are Temperature Distribution, Real-time Fire Information, Fire Risk Evaluation, Facilities Control Panel, and Recommended Actions, as demonstrated in Fig. 11. The Temperature Distribution function is used to display the discrete temperature data along the tunnel measured by ceiling temperature sensors. The temperature sensor data is also used as the input of the AI model. Once a fire is detected, the Real-time Fire Information window will display the real-time fire scenario information including the location and fire HRR predicted by the AI model.

The Fire Risk Evaluation panel is used to estimate the real-time fire risk information, including the vehicle number, fuel load, and fire probability. The Facilities Control Panel is designed to control the tunnel facilities during an emergency manually. There is also a window in the lower middle of the interface to display the recommended actions. The suggestions will change with real-time fire situations. The viewing angle can be adjusted through the buttons on the interface, or it can be controlled with the mouse and keyboard like a 3D game. Fig. 11 demonstrates two different views, front, main, side, and inside views, respectively. The location of vehicles can also be recognized by cameras and computer vision methods, the layout of vehicles here is a hypothetical scenario only. Fire Risk Evaluation and Facilities Control functions are not demonstrated in the present work, because the system has not yet been deployed in an actual road tunnel. The real-time monitoring of temperature distribution and fire information prediction is demonstrated on the developed platform, as shown in Video S1.

Supplementary video related to this article can be found at https://doi.org/10.1016/j.dibe.2024.100381

5.2. Large-scale tunnel fire tests

After the platform is tested using the simulation data, it is necessary to conduct a real-scale demonstration before the real application. To verify and demonstrate the performance of real-time fire recognition of the proposed system, serval real-scale tunnel fire tests were conducted in

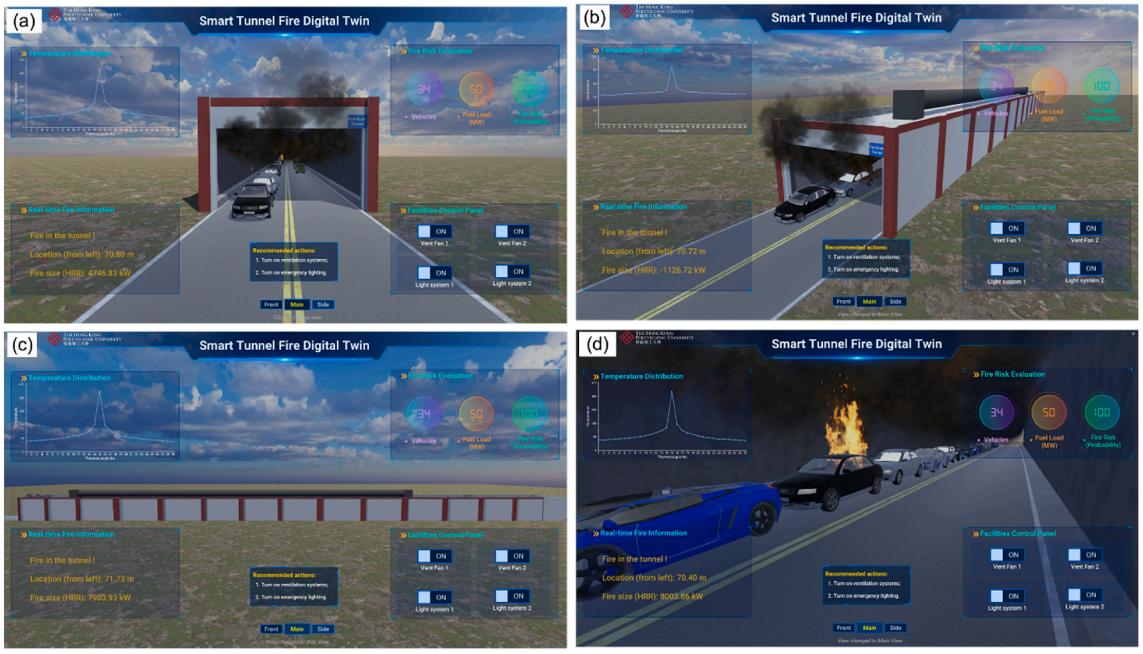


Fig. 11. Demonstration of the developed Smart Tunnel Fire Safety Digital Twin platform.

a large-scale experimental tunnel at SCFRI, which is the basis to build the digital tunnel model.

The tested full-scale tunnel is a rail tunnel with platforms and tracks, and one carriage is parked on the track to simulate the railway tunnel condition, as shown in Fig. 12. Two tunnel portals are connected to the outside environment. The average ambient temperature of the local area was around 20 °C. Inside the tunnel, the extraction ventilation system is installed to exhaust the high temperature and toxic smoke during fire tests. On the tunnel ceiling, there are two smoke curtains to contain the smoke during the fire tests (see Fig. 12). Fireproof materials are adopted to protect the tunnel ceilings and walls during the large fire. As shown in Fig. 12 (b–c), 11 thermocouple trees were used to measure the temperature distribution on the tunnel ceiling. The fire source is placed in the middle of these thermocouple trees. In the current work, only 11 temperature data on the ceiling are used to predict fire information. Two pool fire tests and one wood crib fire test are included and demonstrated in the present study.The temperature distribution of the fire tests at different time are demonstrated in Fig. 12 (d).

For pool fire tests, square fuel pans (0.36 m^2 for case 1 and 1 m^2 for case 2) of different sizes filled with heptane were burned to represent different scales of fire. During the burning, the mass loss rate (MLR) data of the heptane was collected using a scale. Then the HRR values $\dot{Q}$ over time of two tests can be estimated using the following equation:

$$\dot{Q} = \eta \times \dot{m} \times \Delta H \tag{3}$$

where η is the combustion efficiency, the average value of heptane fires is 0.8 in tunnels with natural ventilation (Ishikawa et al., 2021), $\dot{m}$ is the MLR of heptane during the burning (kg/s) and ΔH is the heat of combustion of heptane, which is around 44 MJ/kg (Li et al., 2023). Fig. 13(a and b) demonstrates the profile of HRR over time of the two conducted large-scale fire tests, which was represented by the black dashed line.

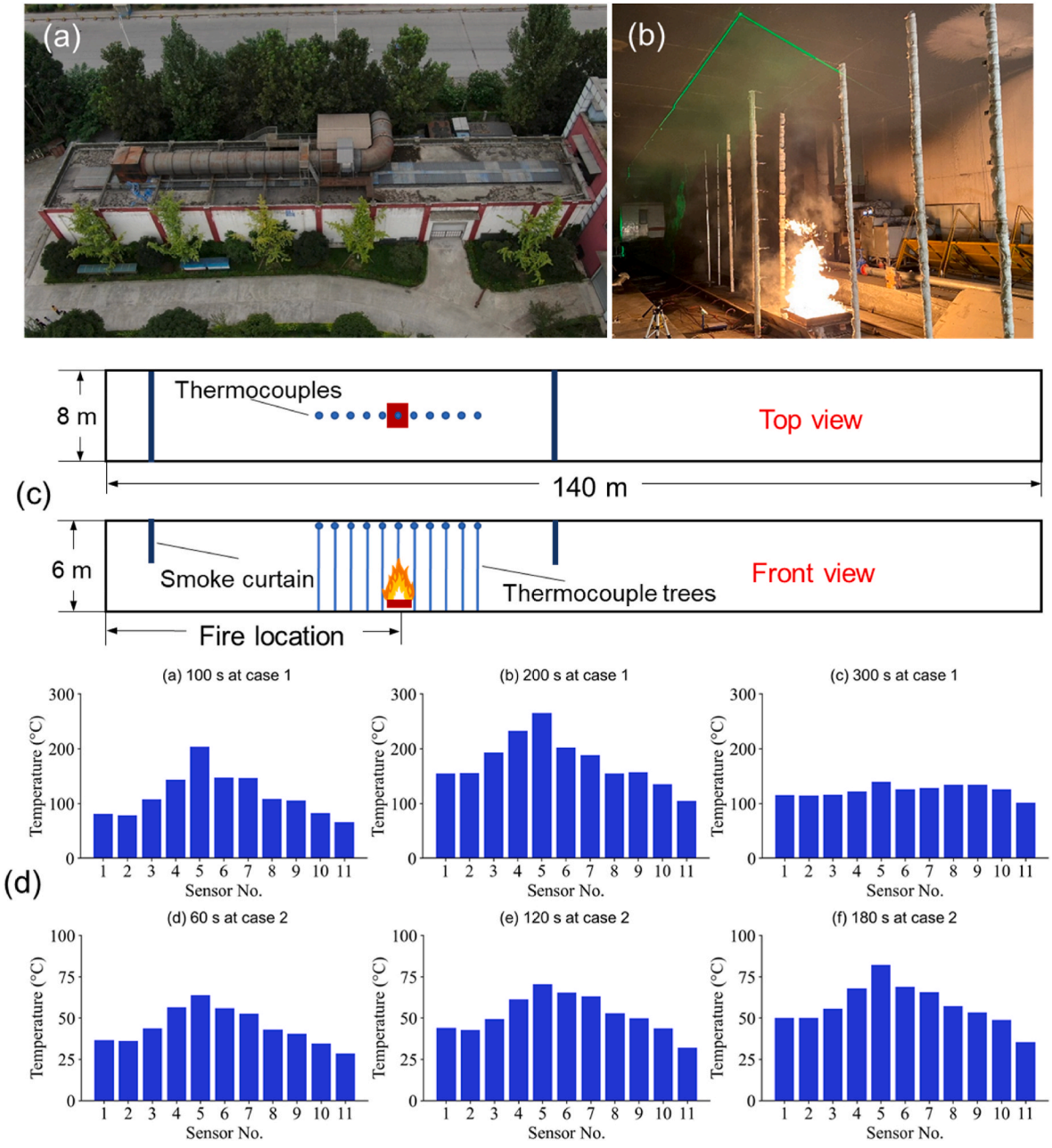


Fig. 12. Systematic diagram of full-scale fire tests, (a) overview of the tunnel, (b) sensor distribution and pool fire, (c) tunnel dimension, and (d) temperature distribution in different case at different times.

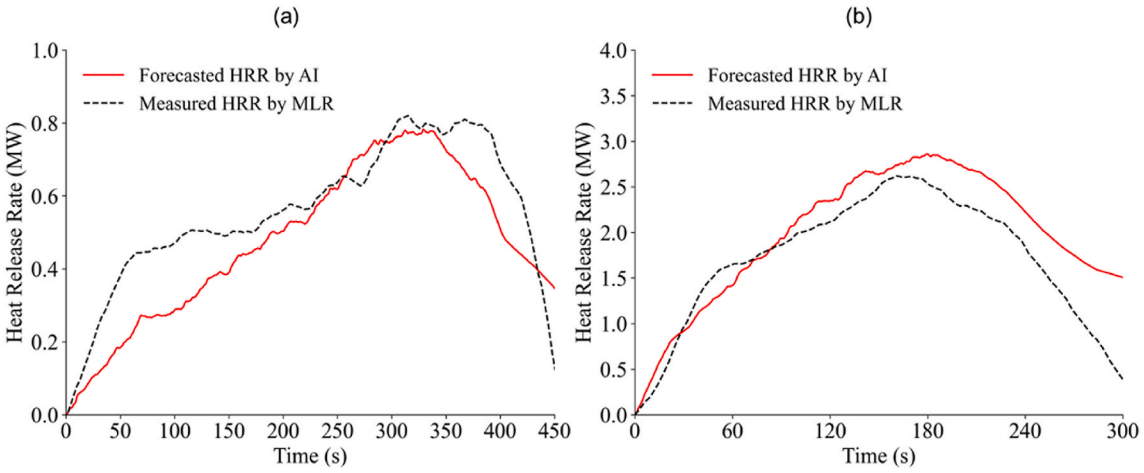


Fig. 13. AI forecasted and measured heat release rate profile in (a) case 1, and (b) case 2.

Based on the conditions of the full-scale fire tests, a new AI model is trained using the numerical simulation data and the proposed Transformer architecture. After training, the model performs similarly to the model for numerical experiments. After the model is optimized and verified, discrete temperature data can be collected and fed into the trained AI model during the real fire tests. The forecasted HRRs by the trained AI model and the measured HRRs by MLR data are compared and demonstrated in Fig. 13, which are represented by red lines. As shown in Fig. 13, the forecasted HRR and measured HRR for both cases demonstrate good performance in fire development and peak predictions.

The wood crib fire test demonstrates the capability of the proposed digital twin in visualizing realistic fire scenarios. The setup and experimental conditions are the same as in previous work (Nan et al., 2023). Three wood cribs were used for the tests, two identical (1.5 m × 1 m × 0.2 m) and one different (1.5 m × 1 m × 0.4 m). The same thermocouple arrangement as in Fig. 12 was applied to record the ceiling temperature profile. The entire test lasted about 140 min, from ignition to extinguishing, and the fire spread from the left wood crib to the right crib. The ceiling temperature profile was input into the trained AI model to predict the real-time fire size, as shown in Fig. 14(b). Serval key time points during the test are analyzed, as displayed in Fig. 14(a and b).

In the beginning, the first wood crib was ignited using wood sticks soaked with heptane. The ignitor accelerated the burning rate and fire spread rate, and the fire size grew rapidly to the fire peak of about 0.8 MW at 15 min. Then the fire size and flame spread rate decreased due to ignitor burnout from 15 min to 58 min. At around 58 min, a greater drop in fire size occurred due to the opening of the ventilation. The third wood crib was ignited at around 80 min, and the fire size began to increase gradually. The fire size eventually peaked at around 1.5 MW in about 100 min and began a decreasing stage at 115 min. The HRR calculated from MLR for a 0.2 m high wood crib is about 0.7 kW, which is in good agreement with the predicted value of the AI model, as shown in Fig. 14(b). The AI model also gives a reasonable prediction of the fire load of the 0.4 m high wood crib, which should be twice the value of a 0.2 m high wood crib. The visualization of the wood crib fire test and comparison with the real fire test are demonstrated in the digital twin platform, as shown in Fig. 14(c) and Video S2.

Supplementary video related to this article can be found at https://doi.org/10.1016/j.dibe.2024.100381

Although the digital twin framework has been successfully demonstrated for a real-scale tunnel with a 3D digital model, there are some issues that should be addressed before the system can be applied to real tunnels. The first issue is the balance between the communication reliability of the IoT sensor network and its cost. In a long road tunnel, using a distributed wireless sensor network (WSN) with Low Power Wide Area Networking (LPWAN) technology to monitor the environment is cheap and easy to install. However, the data capacity and reliability of this kind of sensor are low. During real fire tests, the performance of wireless sensors always fails to meet expectations. Situations such as data loss, data delays, failure of certain sensors, etc., occurred frequently. Possible reasons include the loss of signal due to the narrow and closed tunnel and the damage to the sensor due to the high temperature of the fire and smoke.

As a comparison, traditional wired sensors are expensive but have better reliability than wireless sensors. It may be more suitable for use in fire and other scenarios that require high reliability. Besides, the function of the system in non-fire scenarios should be more precisely defined and developed because there will be no fire in the tunnel in most cases. One of the directions is to improve it into an intelligent tunnel safety monitoring digital twin system that can handle different hazard scenarios. The system can monitor and evaluate risks in everyday situations, visualize onsite scenarios, and provide remote operation interfaces in emergency situations.

The framework proposed in this study is intended to be applied to a variety of tunnels, including those with different shapes, geometries, and ventilation patterns. The core components of the framework, including the data management method, fire safety management strategies, real-time fire and smoke modelling approach and prediction model architecture can be easily adapted for the other tunnels. However, the fire scenario prediction model developed in this study is only adaptable for the tunnel in the experiment conducted in SCFRI. There are some potential methods to solve this problem. The first is to train a new model considering the new tunnel information, but it's extremely time-consuming. The second approach is to train a more general prediction model that can cope with different tunnel shapes and geometries, but such a model requires large amounts of data and computing resources. The last soluction is to adopt transfer learning by using the foundational model trained on the original tunnel and fine-tuning it with a smaller dataset from the new tunnel, which could significantly reduce the model training time.

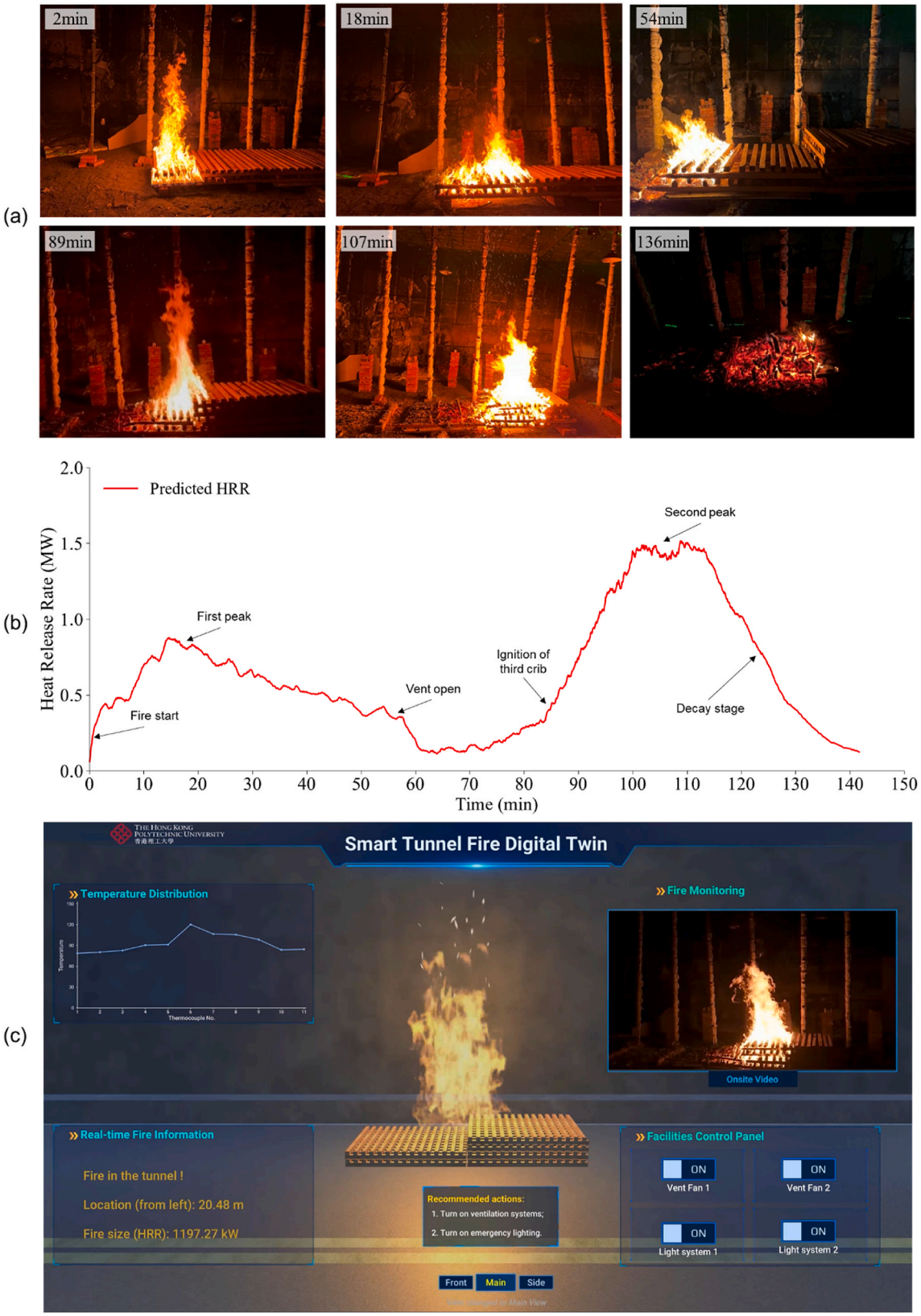


Fig. 14. Demonstration of wood crib fire tests: (a) real fire scene at different key time points, (b) predicted fire HRR of the whole test, and (c) fire scene at tunnel digital twin.

6. Conclusions and future work

In this paper, a digital twin-based smart tunnel fire safety management framework and some implementations of the proposed framework are presented. An AIoT-based smart digital twin framework is proposed and demonstrated for tunnel fire safety management. To enable the fire recognition and reconstruction of the proposed framework, we developed a real-time fire identification model based on deep learning algorithms and numerical simulations. An immersive fire monitoring digital twin platform is developed in Unity 3D, which can visualize the real-time fire scenario and provide a human-computer interaction interface. The accuracy of the established AI model and the capacity of the system are then demonstrated using numerical experiments and large-scale tunnel fire tests. The results demonstrate that the proposed framework and developed prediction model can work well in simulated and full-scale tunnel test scenarios. There are some potential limitations in this study, including the lack of real world data and realistic considerations of the proposed framework, the insufficient assessment of sensor networks and prediction model, adaptability of the proposed framework in different scenarios. These issues may be addressed by collecteing more comprehensive and high quality realistic tunnel fire datasets, leveraging advanced modelling techniques, and conduct comprehensive testing and validation.

The contributions of this study include: (1) a digital twin framework is built up that achieves smart real-time tunnel fire monitoring and fire safety management. (2) validated the performance of AI prediction model using large-scale tunnel fire tests. (3) a platform is built up to visualize the tunnel fire in a 3D environment. Future research can be carried out on the following topics. First, the application of AR/VR in road tunnels to support daily fire safety management and emergency operations, such as the facilities maintenance and operations. Second, the vehicle monitoring and quantitative fire risk assessment inside the tunnel, such as the fuel load distribution and fire probability estimation. Third, the application of robotics in tunnel fire safety management and firefighting operations can reduce the cost of human resources and respond quicker than firefighters.

CRediT authorship contribution statement

Xiaoning Zhang: Investigation, Methodology, Writing – original draft, Software, Validation. **Yishuo Jiang:** Formal analysis, Investigation, Methodology, Writing – original draft. **Xiqiang Wu:** Conceptualization, Investigation, Methodology, Writing – review & editing. **Zhuojun Nan:** Data curation, Formal analysis, Resources. **Yaqiang Jiang:** Project administration, Resources, Supervision. **Jihao Shi:** Formal analysis, Resources. **Yuxin Zhang:** Data curation, Formal analysis, Resources. **Xinyan Huang:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – review & editing. **George G. Q. Huang:** Methodology, Project administration, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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